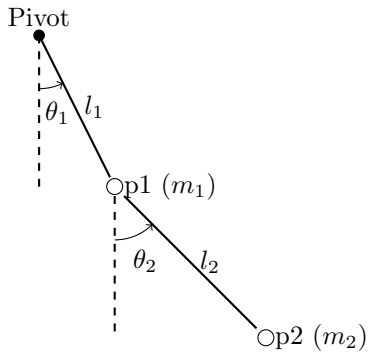
Simulation of a double pendulum system

1 Introduction

So we want to simulate the movement of a double pendulum - how do we do this?
Very briefly, we formulate the Lagrangian or Hamiltonian of the system, then via the action potential and Hamilton's principle of stationary action, transform this into an ODE via the Euler-Lagrange equation, which describes how the system evolves over the time. We can then simulate this via numerical integration with ODE solvers so we can render this in real time.

2 Lagrangian of the System



We can assume the moment of inertia for a simple pendulum with a point mass as $I = mr^2$.
Equations for impulse, kinetic energy and the general Lagrangian of a non-relativistic system are as follows:

$$KE_\omega = \frac{1}{2}mr^2\omega^2, L = T - V, g = 9.81m/s^2$$

Using polar coordinates:

$$p_1 = (l_1, \theta_1), p_2 = (l_2, \theta_2) \tag{1}$$

Our holonomic constraints are:

$$r_1 = l_1, r_2 = l_2 \tag{2}$$

The cartesian coordinates are:

$$p1 = l_1(\sin\theta_1, \cos\theta_1) \tag{3}$$
$$p2 = p_1 + l_2(\sin\theta_2, \cos\theta_2) \tag{4}$$

We will use the following trigonometric identities:

$$\sin^2\theta + \cos^2\theta = 1 \tag{5}$$
$$\cos(\theta_1 - \theta_2) = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 \tag{6}$$

derivations of velocity ($\dot{p_x}$) for $\dot{p_1}, \dot{p_2}$ in cartesian space:

$$\dot{p_1} = \frac{dp_1}{dt} = \frac{dp_1}{d\theta}\frac{d\theta}{dt}, \dot{p_2} = ... \tag{7}$$

$$\therefore \quad \dot{p_1} = l_1\dot{\theta}_1(\cos\theta_1, -\sin\theta_1) \tag{8}$$

$$p_2 = (l_1\sin\theta_1 + l_2\sin\theta_2, l_1\cos\theta_1 + l_2\cos\theta_2) \tag{9}$$

$$\dot{p_2} = (l_1\dot{\theta}_1\cos\theta_1 + l_2\dot{\theta}_2\cos\theta_2, -l_1\dot{\theta}_1\sin\theta_1 - l_2\dot{\theta}_2\sin\theta_2) \tag{10}$$

$$\dot{p_1}^2 = l_1^2\dot{\theta}_1^2 \tag{11}$$

$$\dot{p}_{2,x}^2 = (l_1\dot{\theta}_1)^2\cos^2\theta_1 + 2l_1\dot{\theta}_1 l_2\dot{\theta}_2\cos\theta_1\cos\theta_2 + (l_2\dot{\theta}_2)^2\cos^2\theta_2 \tag{12}$$

$$\dot{p}_{2,y}^2 = (l_1\dot{\theta}_1)^2\sin^2\theta_1 + 2l_1\dot{\theta}_1 l_2\dot{\theta}_2\sin\theta_1\sin\theta_2 + (l_2\dot{\theta}_2)^2\sin^2\theta_2 \tag{13}$$

$$\dot{p_2}^2 = (l_1\dot{\theta}_1)^2 + (l_2\dot{\theta}_2)^2 + 2l_1\dot{\theta}_1 l_2\dot{\theta}_2\cos(\theta_1 - \theta_2) \tag{14}$$

Thus, the formulae for the kinetic energies are as follows:

$$T_1 = \frac{1}{2}m_1 l_1^2\dot{\theta}_1^2 \tag{15}$$

$$T_2 = \frac{1}{2}m_2((l_1\dot{\theta}_1)^2 + (l_2\dot{\theta}_2)^2 + 2l_1\dot{\theta}_1 l_2\dot{\theta}_2\cos(\theta_1 - \theta_2)) \tag{16}$$

$$T = \frac{1}{2}(m_1 + m_2)(l_1\dot{\theta}_1)^2 + \frac{1}{2}m_2((l_2\dot{\theta}_2)^2 + 2l_1\dot{\theta}_1 l_2\dot{\theta}_2\cos(\theta_1 - \theta_2)) \tag{17}$$

Gravitational Potential Energy, V, is as follows:

$$V_1 = gm_1 l_1\cos\theta_1 \tag{18}$$

$$V_2 = gm_2(l_1\cos\theta_1 + l_2\cos\theta_2) \tag{19}$$

$$V = g(l_1(m_1 + m_2)\cos\theta_1 + l_2 m_2\cos\theta_2) \tag{20}$$

$$\tag{21}$$

The equation for the Lagrangian of the system is as follows:

$$L = \frac{1}{2}(m_1 + m_2)(l_1\dot{\theta}_1)^2 + \frac{1}{2}m_2((l_2\dot{\theta}_2)^2 + 2l_1\dot{\theta}_1 l_2\dot{\theta}_2\cos(\theta_1 - \theta_2)) - g(l_1(m_1 + m_2)\cos\theta_1 + l_2 m_2\cos\theta_2) \tag{22}$$

3 Euler-Lagrange Formulation

The lagrangian is a function of the form $L(t, \theta, \dot{\theta})$, and the action function is defined as the integral of the lagrangian: $\int L(t, \theta, \dot{\theta})dt$. From Hamilton's principle of stationary action and the Euler-Lagrange differential equation, we can say that the system evolves as following:

$$\frac{\partial L}{\partial\theta} - \frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}} = 0 \tag{23}$$

So from here we *'just'* throw in our equation for the Lagrangian and solve using a numerical integrator, right? Terms for θ_1:

$$\frac{\partial L}{\partial\theta_1} = -m_2 l_1\dot{\theta}_1 l_2\dot{\theta}_2\sin(\theta_1 - \theta_2) + gl_1(m_1 + m_2)\sin\theta_1 \tag{24}$$

$$\frac{\partial L}{\partial\dot{\theta}_1} = (m_1 + m_2)l_1^2\dot{\theta}_1 + m_2 l_1 l_2\dot{\theta}_2\cos(\theta_1 - \theta_2) \tag{25}$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}_1} = (m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) - m_2 l_1 l_2\dot{\theta}_2\sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2) \tag{26}$$

$$\tag{27}$$

$$\begin{aligned}\therefore \quad & -m_2 l_1\dot{\theta}_1 l_2\dot{\theta}_2\sin(\theta_1 - \theta_2) + gl_1(m_1 + m_2)\sin\theta_1 \\ & -((m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2 l_1 l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) - m_2 l_1 l_2\dot{\theta}_2\sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)) = 0\end{aligned} \tag{28}$$

Terms for θ_2:

$$\frac{\partial L}{\partial \theta_2} = m_2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + g l_2 m_2 \sin\theta_2 \tag{29}$$

$$\frac{\partial L}{\partial \dot{\theta}_2} = m_2 l_2^2 \dot{\theta}_2 + m_2 l_1 \dot{\theta}_1 l_2 \cos(\theta_1 - \theta_2) \tag{30}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_2} = m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 \ddot{\theta}_1 l_2 \cos(\theta_1 - \theta_2) - m_2 l_1 \dot{\theta}_1 l_2 \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2) \tag{31}$$

$$\begin{aligned} \therefore \quad & m_2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + g l_2 m_2 \sin\theta_2 \\ & - (m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 \ddot{\theta}_1 l_2 \cos(\theta_1 - \theta_2) - m_2 l_1 \dot{\theta}_1 l_2 \sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2)) = 0 \end{aligned} \tag{32}$$

4 ODE Solvers

4.1 Verlet

It would be preferable to use Verlet over RK4, as RK4 requires converting our second order ODE's to a system of first order ODE's, and the Velocity Verlet variant works with second order ODE's. Verlet is 'sympletic' (no, I don't know how to explain that! Ask your local maths grad). It basically has better stability than RK4. Normal Verlet integration is given by:

$$\theta_{i+1} = \theta_i + (\theta_i - \theta_{i-1}) + a \cdot dt_i^2 \tag{33}$$

However, verlet breaks down quite badly when the time step is not consistent, luckily, we have:

- time-adjusted verlet,
- and a potiential improvement

We will use the following formula for time-corrected Verlet:

$$\theta_{i+1} = \theta_i + (\theta_i - \theta_{i-1}) \cdot \frac{dt_i}{dt_{i-1}} + a \cdot dt_i \cdot \frac{dt_i + dt_{i-1}}{2} \tag{34}$$

It may also be worth looking at other integrators in the future such as:

- Beeman's algorithm
- Velocity Verlet

4.2 Runge-Kutta-Nyström Methods

NASA TR R-432 is a useful source on Runge-Kutta Nyström Methods.
Runge-Kutta-Nyström Methods are well suited to our use case of having a second order ODE.

4.3 RK4

This will require transforming our system of second-order ODE's to a system of first-order ODE's. given our ODE in the form:

$$\text{given} \quad \dot{\theta} = f(t, \theta), \quad \text{step size } h > 0, \text{time steps} \quad t_{n+1} = t_n + h, \tag{35}$$

$$\theta_{n+1} = \theta_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{36}$$

$$k_1 = f(t_n, \theta_n) \tag{37}$$

$$k_2 = f(t_n + \frac{h}{2}, \theta_n + k_1\frac{h}{2}) \tag{38}$$

$$k_3 = f(t_n + \frac{h}{2}, \theta_n + k_2\frac{h}{2}) \tag{39}$$

$$k_4 = f(t_n + h, \theta_n + hk_3) \tag{40}$$

$$\tag{41}$$